

SHIVA HARI GUNDETI

☎ (801) 634-8493 ✉ shivaharigundeti@gmail.com 🔗 [linkedin.com/in/gundetishivahari](https://www.linkedin.com/in/gundetishivahari) 🌐 shivaharisonu.github.io

SUMMARY

Software Engineer with over 3.5 years of experience in Fullstack and Data Platform teams. Proficient in Python and Java with an expertise in large scale data management and building backend scalable systems. Researcher in the realm of Database systems, distributed systems with an expertise in writing algorithms and training models, and data preprocessing.

TECHNICAL SKILLS

- **Programming:** Python, Java, C++, HTML/CSS, Javascript, SQL, PySpark
- **Machine Learning:** PyTorch, Tensorflow, Scikit-learn, Matplotlib, Keras, Nltk, OpenCV, SpaCy
- **Full Stack development:** Spring-Boot, Django, React, Typescript, Node.js, GIT, Docker, Kubernetes, CI/CD
- **Technologies:** Kafka, Flink, Neo4J, Airflow, Elasticsearch, Apache Spark, MongoDB, AWS S3, DynamoDB, Redis

PROFESSIONAL EXPERIENCE

Apple

Software Engineer (ICT 3)

[Oct'25-Present]

Austin, Texas, USA

- Created a full stack batch data transformation application to perform Retail Store Replenishment calculations using Gurobi Solver and SingleStoreDB.
- Migrated the application from the old stack Oracle PL/SQL to Polars for small scale data transformations and Spark for large scale data transformations. Built Airflow dags to schedule the ETL jobs from AWS S3 to SingleStore tables.
- Developed AI agents that support OIDC authorization for chatbot with MCP tools, trigger alerts from anomalies.
- **Technologies:-** Python, React, SingleStore DB, Airflow, Docker, Gurobi Solver, Polars, Pyspark

University of Utah Health

Graduate Research Assistant

[Jan'25 - June'25]

Salt Lake City, Utah, USA

- Created a full stack application to visualize the **time series forecasts** of diseases with interactive geomaps using Plotly.
- Built **Kafka pipeline** that batches weekly data and triggers forecasting models (**Chronos, ARIMA, VAR, Transformer**)
- Integrated streaming data with historical database records for model training, storing predictions in **PostgreSQL**
- Deployed it on Google Cloud Run in a Dockerized environment, integrating automated CI/CD pipelines via GitHub Actions.
- **Technologies:-** Python, Django, React, PostgreSQL, Apache Kafka, Plotly, Darts

Seismograph stations, University of Utah

Student Research Assistant

[Feb'24 - Aug'24]

Salt Lake City, Utah, USA

- Trained **CNN and LSTM** models to perform spectrogram classification on mining signals and attained 95.8% accuracy
- Researched weight initialization techniques (**Xavier, He, Normal**) and activation functions to mitigate gradient problems.
- Improved the robustness by implementing **SWAG** algorithm and achieved a **97.3%** test accuracy on **CNN architecture**.
- Performed **Multi-SWAG Inference** using ensemble of **CNN, LeNet, and PreResNet** to improve on calibration plot.

Brane Enterprises

Software Development Engineer 2

[April'23-June'23]

Hyderabad, Telangana, India

- **Data Indexing and Ingestion:-** Built **indexing service** to create **ETL pipelines** using **Apache Flink** to process and stream data in real-time to **Elasticsearch** resulting in highly available pipeline with a p99 latency of 178 milli seconds.
- **Search Analytics:-** Architected **OLAP pipelines** with Kafka and Flink to capture user clicks to **AWS S3**. Integrated with **Apache Superset** using Athena plugins for dashboard analytics.
- Developed a POC to preprocess data to extract semantic words by training the **Word2Vec model** by the **Continuous Bag of Words (CBoW)** algorithm. Created **knowledge graphs** on Neo4J using similar words for semantic search.
- **E-Commerce based search:-** Developed search application supporting faceted search and typeahead. Leveraged Elasticsearch to search 60M+ records with 80-180ms latency for multi-tenant, multi-language with user configurations.
- **Notifications:-** Developed Kafka notification system with **AWS SNS, SQS, and Airflow** for scheduled messaging handling 10k notifications per day.
- **Auditing and Metering:-** Implemented Aspect-Oriented Programming (AOP) to enhance REST APIs with auditing and metering, integrating with 50+ microservices and streaming API usage data via Kafka to Druid for real-time analytics.
- **Technologies:-** Java, Python, Spring-boot, Apache Druid, Apache Kafka, MongoDB, Elasticsearch, Airflow, AWS S3, Neo4J, Apache Flink, AWS Athena, Apache Superset

EDUCATION

University of Utah

Master of Science in Computer Science - GPA: 3.95

Utah, USA

[Aug'23-May'25]

Related Courses: Graduate Algorithms, Deep Learning, Natural Language Processing, Manage Data with Machine Learning

Indian Institute of Technology

Bachelor of Technology - Electrical Engineering - GPA: 7.97

Bhubaneswar, India

[Jul'17-May'21]

Related Courses: Software Engineering, Software Testing and Verification, Machine Learning and Data Analytics